

Initial Calculations

INA219 — Pre-build theoretical values and configuration parameters

Project	Battery-Charge-Controller	Branch	current-sensing
Document		Date prepared	July 2026 (pre-build)
Sensor	INA219 breakout module (0.1Ω shunt, I2C)	Status	THEORETICAL — awaiting hardware verification

Purpose: All values here are calculated from datasheets and theory. When real hardware is connected, Discrepancies between theoretical and measured are documented as engineering observations.

1. INA219 Key Specifications (From Datasheet)

Parameter	Value	Notes / Source
Supply voltage (VCC)	3.0 V – 5.5 V	Connect to Arduino 5V rail
Max bus voltage (VIN+)	26 V absolute max	Well above your 7.3V battery — safe
Shunt resistor (module)	0.1 Ω, 1% precision	Pre-soldered on breakout board
Max shunt voltage range	±320 mV (default PGA ÷8 setting)	= 3.2A max at 0.1Ω
Current range (default)	±3.2 A (bidirectional)	More than enough for 0–2A charge current
Current resolution (32V/2A)	0.1 mA per LSB	Adafruit library default calibration
ADC resolution	12-bit (4096 steps)	Shunt voltage register
Shunt voltage LSB	10 μV	= 40mV / 4096 at ÷1 gain
Bus voltage LSB	4 mV	= 16V / 4096
I2C address (default)	0x40 (A0=GND, A1=GND)	Do not bridge address pads
I2C pins — Arduino Mega	SDA = Pin 20, SCL = Pin 21	Hardware I2C — do not use other pins
Pull-up resistors	4.7 kΩ on SDA and SCL to 5V	Usually pre-installed on breakout module
Bypass capacitor	100 nF ceramic on VCC pin to GND	TI datasheet recommendation

Table 1: INA219 specifications from TI datasheet

2. Calibration Register Calculation

2.1 Design Target

The charge controller charges at a maximum current of 2.0A (0.5C for a 4Ah pack, or 1C for a 2Ah pack). The INA219 must resolve this accurately enough for the Coulomb counting SoC algorithm — target resolution is better than 5 mA.

2.2 Current LSB Calculation

$$\begin{aligned}\text{Current_LSB_max} &= I_{\text{max}} / 2^{15} = 2.0 \text{ A} / 32768 = 61.0 \mu\text{A} \\ \text{Chosen Current_LSB} &= 61 \mu\text{A} \text{ (nearest value, gives best resolution)}\end{aligned}$$

Each count in the INA219 Current register represents 61 μA of actual current. At 2.0A charging current, the register value will be:

$$\text{Register value at 2A} = 2,000,000 \mu\text{A} / 61 \mu\text{A} = 32,787 \text{ counts}$$

This is within the 16-bit signed range (max 32,767 positive counts) — marginally. In practice the Adafruit library uses a Current_LSB of 100 μA (setCalibration_32V_2A), which gives a cleaner resolution of 0.1 mA and stays well within the register range at 2A (20,000 counts).

2.3 Calibration Register Value

$$\begin{aligned}\text{CAL} &= \text{trunc}(0.04096 / (\text{Current_LSB} \times R_{\text{shunt}})) \\ \text{Using Current_LSB} &= 100 \mu\text{A} \text{ (Adafruit 32V/2A default):} \\ \text{CAL} &= \text{trunc}(0.04096 / (0.0001 \times 0.1)) \\ \text{CAL} &= \text{trunc}(0.04096 / 0.00001) = \text{trunc}(4096) = 4096 = 0x1000\end{aligned}$$

The Adafruit library writes 0x1000 to register 0x05 automatically when you call setCalibration_32V_2A(). This is the value you should see if you read back the Calibration register during debugging.

2.4 Verification: Expected Shunt Voltage at Target Currents

Charging Current	Expected V_shunt	Register value ($\div 8$ PGA)	INA219 reports (mA)
0.0 A (idle)	0.0 mV	0	0.0 mA
0.5 A (0.5C for 1Ah)	50.0 mV	500	500.0 mA
1.0 A (1C for 1Ah)	100.0 mV	1000	1000.0 mA
1.5 A	150.0 mV	1500	1500.0 mA

2.0 A (maximum)	200.0 mV	2000	2000.0 mA
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Table 2: Expected INA219 readings at key current levels (theoretical)

3. Expected Simulation Readings (9V Supply, 10Ω Load)

In the Proteus simulation, a 9V source feeds through the INA219 shunt (0.1Ω) and into a 10Ω test load resistor. Expected values:

$$\text{Total resistance} = R_{\text{shunt}} + R_{\text{load}} = 0.1 + 10.0 = 10.1 \Omega$$

$$\text{Expected current} = 9.0 / 10.1 = 0.8911 \text{ A} = 891.1 \text{ mA}$$

$$V_{\text{shunt expected}} = 0.8911 \times 0.1 = 89.1 \text{ mV}$$

$$\text{Bus voltage (at VIN-)} = 9.0 - 0.0891 = 8.911 \text{ V}$$

$$\text{Power (at load)} = 8.911 \times 0.8911 = 7.942 \text{ W} = 7,942 \text{ mW}$$

Parameter	Theoretical Value	Acceptance Range	Simulation Reading
Current	891.1 mA	880 – 905 mA (±1.5%)	<u>891.10</u> mA
Shunt voltage	89.1 mV	88 – 91 mV	<u>89.11</u> mV
Bus voltage	8.911 V	8.85 – 8.97 V	<u>8.912</u> V
Load voltage	9.000 V (= bus + shunt/1000)	8.96 – 9.04 V	<u>9.001</u> V
Power	7,942 mW	7,750 – 8,150 mW	<u>7940</u> mW

Table 3: Theoretical vs simulation results